

Disc 20

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0.1 1

a. If A is a $n \times n$ symmetric matrix, then $Ax = xA$

True, even though multiplication is not commutative, our $A^T = A$ is, meaning it will hold? $Ax = (Ax)^T x = x^T A^T x = x^T A x = xAx$

b. If A is a $n \times n$ symmetric matrix, and B is a $n \times m$ matrix, then $B^T A B, B^T B, B B^T$ are all symmetric. The approach I chose was using the dimensions.

$$B^T = (m \times n)$$

$$B^T A = (m \times n), (B^T A) B = (m \times m) \implies \text{symmetric}$$

$$B^T B = (m \times n)(n \times m) = (m \times m) \implies \text{symmetric}$$

Approach is to transpose it and show its equal e.g.

$$(B^T A B)^T = \text{the start}$$

c. Let $u \in \mathbb{R}^n$ be a unit vector. Let $C = uu^T$

1. Show that Cx is the projection of x onto u

2. Compute the eigenvalues of C

3. Show C is symmetric and $C^2 = C$

1. $proj_u(x) = (u^T x)u = Cx = u(u^T x)$ 2. The image of C is a 1d subspace, and therefore we only have to find one.

$$Cu = u(u^T u) = u$$

3. $C^T = (uu^T)^T = \dots = u(u^T u) = u$

0.2 2

If A is a $n \times n$ invertible symmetric matrix s.t. quadratic form $x^T A x$ is positive definite, then all quadratic form $x^T A^{-1} x$ is also positive definite

$$x^T A x = x^T P^T D P x = y^T D y = y_1 \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2 > 0$$

Definitely positive, comments were that it means nothing. We just need to show that $x^T A x > 0$

The basis of x will always be positive implies that y is also positive, likewise for y to x . Positive matrices derive to positive ???

$$A = P^T D P = A^{-1} = P^T D^{-1} P$$

0.3 3

Show that if A is square, then

$$|\det(A)|$$

is the product of the singular values of A

nonsquare matrices also have something LIKE eigenvectors and eigenvalues, by singular values.

Its still useful to find decomposition of matrices even when they are not square. Orthogonal matrices are just rotations $A = U \Sigma V^T$, A is always in the form of rotate, scale, rotate

Take the det of both sides to show that both sides are 1.

$$A = U \Sigma V^T = \det(A) = \det(U \Sigma V^T) = \dots = \det(\Sigma) = \pm \pi \sigma, \text{ where } \sigma = \text{singular value}$$